

Academic Integrity Promotion & Prevention Plan

Introduction to R Programming

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Plan Overview

I will teach learning skills, AI competencies, and use a specs grading system in my Introduction to R Programming course.

Learning Skills Overview

Obstacles to learning make paths of least resistance appealing. I will promote academic integrity through value-based activities completely integrated into the course design. In my course, I will shape learning skill behavior alongside statistical coding skills. Student work is evaluated for the use of learning skills through the rubrics. Most of the learning skill vocabulary terms that I highlight are mostly from Jason Baehr's book *Deep in thought: A practical guide to teaching for intellectual virtues*. Baehr calls these terms intellectual virtues. I chose to use the term learning skills because it seemed more appropriate for grading. I suppose I am concerned that students might take my feedback negatively if they do not meet rubric expectations, if I use the term 'virtues'. Using the term 'learning skill' feels better to me when I am determining whether the student is displaying this skill.

My motivation for utilizing this approach is aligned with Baehr's three ideals he lists in his first chapter: academic, social, and economic and professional. Academic ideals might be considered what the student will be able to do at the end of the course —the CLOs. For the Introduction to R Programming course, the three main CLOs are (1) Share a reproducible R code workflow using the desktop RStudio; (2) Confidently problem solve R coding tasks; and (3) be ready for their next R course. Since learning skills are built into the course design, I include a CLO for them. (4) Demonstrate learning skills & AI competencies in workflow. In my course, the social ideals concern how the student will behave as a student, a future researcher, and a member of our society. My motivation for using learning skills is to (1) protect the integrity of science; (2) protect the integrity of the institution; and (3) protect the value of the degrees awarded by the university. Additionally, I want my teaching practices to equip graduates to make productive contributions to their communities by understanding the skills and values needed to collaborate in resolving current and emerging challenges. Finally, the economic and professional ideals in my course are how a student behaves as a member of the workforce, including (1) collaborating, (2) critically evaluating, (3) communicating, and (4) creating with integrity,

responsibility, professionalism, positive attitude, flexibility, and work ethic. Surely these are dreams that are beyond the scope of any one class. Teaching learning skills to accomplish these ideals may provoke skepticism as to whether I could make such an impact. I don't think that there is any harm in using these ideals to guide my pedagogical practice.

[paragraph about the selected intellectual virtues/learning skills]

Each module highlights one learning skill in the course content. Students will be asked to apply the learning skill to their work in specific ways. In the regular assignment, students will write memos, journaling their learning experience. As the course progresses through the learning skills, the memoing procedures will be shaped into more formal documentation practices commonly seen in researcher journaling. Additional activities are offered throughout the course, such as (1) students reflecting and discussing with their peers about their academic goals and values and (2) readings and discussions about examples of researcher misconduct and how it harms the integrity of science.

Table 1. Intellectual-virtue learning skills with definitions and concrete memo examples. Skills are from Baehr (2022). Transparency is defined by Resnik & Elliott (2023).

Skill	Meaning	Example
Curiosity	Asking new, relevant questions and investigating them beyond what's assigned.	"While completing Q3, I wondered if the trend would look different by gender. I created an extra plot splitting the data by 'gender' to see."
Autonomy	Taking initiative and making choices about your learning path, rather than waiting for exact instructions.	"Instead of using the dataset in the example, I found a similar one from [source] so I could practice the same function on fresh data."
Thoroughness	Clearly documenting your steps and reasoning so someone else can follow your process.	"I tried two different ways to calculate the mean: 'mean()' and 'summarise()'. I noted both in my code with comments and explained why I chose to keep the 'summarise()' method."
Open-mindedness	Considering and testing alternative methods or perspectives, even if you already have a working answer.	"After getting the result with 'lm()', I also tested a robust regression with 'rlm()' to see if outliers were affecting my slope."
Courage	Speaking up with ideas or questions, especially when you're unsure, and sharing your work for feedback.	"In class, I asked my partner to check my code because I wasn't sure if my use of 'mutate()' was efficient. Their suggestion helped me simplify it."
Tenacity	Persisting through difficulties, revising code, and knowing when to seek help after genuine effort.	"It took me three tries to debug the error in Q5. I checked my syntax, then my variable names, and finally Googled the error message before finding the typo."
Transparency	Disclosing methods, materials, assumptions, values, and interests.	"I used boiseestate.ai to figure out the error 'Error in 'mutate()': object 'x' not found" in my code. It was due to a missing package, which I installed. I noted this in my code comments."

Teaching learning skills will allow me to promote the development of intellectual character. I want my teaching to promote a love for learning. I recognize that I will not achieve a wide-scale transformation of my students' learning practices. That does not make these goals any less worthwhile.

AI Competencies Overview

One path of least resistance is to use artificial intelligence (AI) to generate code for students. The rapid proliferation of AI across all aspects of life is posing challenges to education systems aimed at preparing students. Given the transformative potential of AI for human societies, it is crucial to equip students with the values, knowledge and skills needed to navigate the future with AI. Students using AI will undoubtedly have an impact on their learning. I believe anecdotally that using AI facilitates learning. It can also prevent it. It all depends on how it is used, the specific use case. Along with targeting learning skills, I will teach several AI competencies as they apply to this course. The AI competencies, as defined by Miao & Shiohira (2024), include:

1. Human-centered mindset: The dynamics of humans controlling AI versus AI controlling humans.
2. AI techniques and applications: Whether or not to use a specific AI system to achieve a justified aim. Before using AI, students will ask:
 - a. Should AI be used in a particular situation?
 - b. Which AI tool to select for a given purpose or task?
 - c. What are the boundaries, goals, and constraints of a problem?
3. Ethics of AI: Transparency of use and guidelines for utilizing AI within rules and regulations, and privacy risks of certain AI systems.

In each module, students will be provided with research related to AI competency development as it relates to that module's learning skill and/or R programming topic.

Specs Grading

This course will use a specifications grading system. This "specs grading" raises standards, motivations and clarity while reducing stress and building career competencies (Nilson & Packowski, 2025). Simply put, this means students can select the grade they wish to earn in the course. Grades will be earned by completing course content bundles. All work is evaluated as "complete," "not yet" (revision required), or "missing" language. In other words, each assignment or activity has a pass/fail rubric. There will not be partial credit. However, students will receive three revision tokens at the beginning of the course and can earn up to three more tokens by completing additional tasks or doing exceptional work. Tokens allow students to revise "not yet" or "missing" items.

Students will have an at-a-glance grade map that follows the core specs of one-level rubrics. The course will have a consistent module pattern for seven modules. Fourteen core assignments, two per module, are required to complete the course at a D grade. To meet the complete specs for core assignments, the submission requires all elements to be true:

- Scope: all questions are answered.
- Correctness: Code runs without errors and fulfills the prompt tasks.

- Clarity: Code documentation file is readable with formatting
- Correct: file code documentation and data are submitted on time.

Successful completion of the core assignments and one basic signature assignment earns students a C grade.

Students wishing to earn a higher grade will select activities and assignments from a menu board within each module, as well as additional signature assignment options for a B or A grade. With a more advanced signature assignment (or two basic), completing x number of menu board activities will earn a B and completing y number of activities will earn an A, where $x < y$. This stepwise bundle design is simple and transparent.

Basic signature assignment options might be:

- Wrangling Portfolio: three messy (supplied) datasets cleaned/tidied scripts + short write-ups.
- Visualization Gallery: 5 purposeful charts (different geoms), each justified for audience/task.
- Personal Help File (enhanced): curated “how I solved it” recipes tied to course topics.

Advanced signature assignment options might be

- Submitting a basic signature assignment with R Markdown or Quarto-designed HTML
- Providing evidence of linking the computer to GitHub within the basic signature assignment
- Acquiring, citing a real open-source dataset and applying data wrangling and visualization techniques that are logical for the specific data.

To ensure students understand this process, I will have a “How to Earn Your Grade” page and link it in every module. Additionally, I will have students declare which bundle they want to achieve and check in with them throughout the semester to help them achieve their grade goal.

I will provide ample examples of what “complete” and “not yet” work looks like. Additionally, I will need to be very clear about how to save and upload the correct files. Students have significantly struggled with correct submissions in the current course design.

Seven Module Breakdown

I use icons to help facilitate the learning of how to demonstrate learning skills. These icons will connect the examples I provide in the lesson videos, example documents and rubric checklist. Each module will build on the next, beginning with a simple, disorganized, memoing techniques that look more like running trains of thought than academic work. As modules progress, memos will become more structured until they form a tidy internal R report document fit for sharing with research collaborators for reproducible, transparent research. Additional learning skills and AI competency activities are provided in modules for students seeking higher course grades.

Module One



Tenacity — Students today are finding it difficult to persevere through errors in code while learning this foreign R coding language in the Introduction to R Programming course. We know that tenacity is what skilled learners use to achieve [citation?]. It is easier now than ever to hand off tasks to Large Language Models (LLMs). Without developing grit, student learning will suffer. If these AI tools are very good at producing code, why should a student learn to write the code themselves? Simply put, data analysts need to be able to explain their results. I will present the values associated with the integrity of science, explanatory power, and the AI competency of algorithmic opacity, explaining why learning what the code means is important. But that is not enough. I will need to develop a reward system for using the learning skill tenacity. Students will document their struggles through writing memos with their code. I ask students to comment out the code that did not work and record what they think the problem was. I demonstrate this in my examples, and the rubric will (🎯 Goal and 🔍 Troubleshooting icons introduced). Then the student is to test their idea by modifying the code to see if it works. The code in the first module is especially easy, and most student errors will be due to simple typos. Students in this module who are looking for a higher course grade will have activities that explore why they should learn R.

Module Two



Courage — As the course moves into slightly more challenging coding concepts, students need to know how to find the help they need in ways that promote learning. Fear is an obstacle to learning. Learners may fear failure, being exposed as ignorant or looking foolish, which prevents them from speaking up. The consequences are severe, such as disengagement, missed opportunities, and passive participation. Just last week, I finally reached out to a student who had stopped submitting assignments. The student informed me that they had encountered an error that prevented them from completing their assignment. Out of frustration, this individual stopped working on the assignment and the course altogether. Unfortunately, this student missed the video I had made showing how to overcome this specific problem. Courage is not the absence of fear but the willingness to take the intellectual risk of speaking up. At the beginning of the module, I will show students various ways to troubleshoot their code. In a series of videos, I will demonstrate how to use various resources, including Google, Stack Overflow, the RStudio help section, and the approved LLM (boisestate.ai). The AI Competency focus for this module is AI techniques and applications for learning code. Students will learn how to document their questions, the resources they use, and how they helped, without copying and pasting (👁️ Resource Cited, 🧠 Growth mindset, and ❓ Question icons). A growth mindset is shown in memos through a positive attitude ("Not yet"), asking questions, and seeking help. At the end of the module, students will complete group activities that begin with privately reviewing their feedback and the assignment they submitted, then crafting questions. They will

then ask their peers these questions and work together to solve them. [I will need to make sure to be on top of grading these first few weeks!!]

Module Three



Transparency — A natural next step is to highlight the learning skill of documenting research processes. Transparency is defined by Resnik & Elliott (2023). While this is not an intellectual virtue highlighted by Baehr or King, it is an academic integrity and might be closely associated with honesty. In my course, I use the term transparency because a learning outcome is to create reproducible scientific work using R code. In this module, students will begin to structure student memoing practices (📝 Document code tried, 📄 Formatted document, 🕒 Dated document icons and reinforced 📖 Resource Cited). Students seeking higher grades will have activity options on the Ethics of AI as part of AI Competency development.

Module Four



Curiosity — A healthy appetite for knowledge seeks truth, knowledge, and understanding within the bounds of the student's goals. This value is counter to indifference and gluttony (King, 2021). Indifference is the “I don't care” attitude, not the “I don't know” attitude. Gluttony is when curiosity takes control of the learner, leading them to chase the wrong things at the wrong time or for the wrong reasons (King, 2021). In this week's module, students seeking higher final grades will engage in activities that examine their academic goals and values and discuss them with their peers. Students looking for a higher final course grade will have activities that teach them more details and functionalities of RStudio (🚀 Explore and Play, 🔧 Break Things!, and 🔍 Look Deeper icons, and the reinforced ❓ Question icon). The AI Competency activity will look at a Human-centered mindset and the concern of “parroting” versus authentic, original ideas. Additional discussion and activities will address AI techniques and applications.

Module Five



Autonomy — “Auto” signals that intellectual responsibility lies with the individual. But autonomy requires that we think for ourselves and not by ourselves. Too much autonomy (isolation) can cause students to “reinvent the wheel” (King, 2021). Students will participate in a group activity to select a dataset for their signature assignment. Students seeking higher grades will have options for activities that cover a Human-centered mindset in AI Competency development. This competency will focus on issues with copy-pasting and summarizing learning (💡 Comment and 🔄 Metacognition icons introduced).

Module Six



Openmindedness —

Students looking for a higher final grade will have an activity that compares code that effectively does the same action.

Module Seven



Throughness — In Fyodor Dostoevsky's 1869 novel *The Idiot*, the main protagonist, Prince Myshkin, said, "We can't understand everything at once, we can't begin directly from perfection! To achieve perfection, we must first of all fail to understand a great many things! And if we understand too quickly, we may not understand very well." This learning skill is the balance between superficial understanding and perfectionism. The AI Competency activity will look at a Human-centered mindset and AI techniques and applications to achieve a deeper understanding. The structure of the memo notes is strict for spec grading on signature assignment where students must record their (1) 🎯 Goal where they add comments before or near their code so a collaborator can reproduce their steps; (2) ? Question / 💬 Prediction where they state their question or hypothesis; (3) ⚡ Test where students compose the code to achieve the goal, answer the question and test their prediction; (4) 📝 Document what code they tried, even if it didn't work or achieve the results they were hoping for; ✅ Verify which means to check that results are correct; (5) 🧠 Comment is where the student explains what they learned from the test; and finally (6) 🔄 Metacognition which is where students explain why what they learned will be useful.

Measures of Success

In this section, I will explain how I will conduct SoTL research to evaluate the impact this teaching plan has on student success. More to come soon.

References

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